



TÉCNICO
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Modelling an open water channel

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Outline

- Introduction
- Objectives
- Methodology
- Simulation & Results
- Conclusion



(IEPSL – 2008)

Introduction

- Study of an open water channel system.
- **Focus:** modulation of a water flow using a gate.
- **Theoretical base:** Saint-Venant equations simplified for planar base and constant depth.
- **Tool used:** MATLAB simulation

Motivation

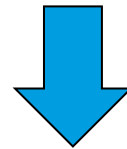
- Understanding water channel behaviour is key for efficient water distribution.
- Helps with the design and control of irrigation and hydraulic systems.
- Basis for future automation and control tasks in open-channel systems.

Objectives

- Model the flow modulation of an open water channel.
- Simulate how water height changes with different gate apertures.
- Identify equilibrium and analyse system stability.
- Compare results from different initial conditions and parameters.

Methodology (1)

$$\frac{dh(t)}{dt} = -\frac{ub\sqrt{2gh(t)}}{A} + \frac{10^{-3}Q}{A}$$



$$x((k+1)\Delta t) = \Delta t \cdot \left(-\frac{ub\sqrt{2gx(k\Delta t)}}{A} + \frac{10^{-3}Q}{A} \right) + x(k\Delta t)$$

1. Use of Euler's Method to obtain a discrete equation
2. Implementation of the equations using MATLAB
3. Perform of the simulations:

Methodology (2)

Simulations performed:

- System response for a gate opening of 50% and 55%
- Different initial conditions (water height)
- Parameter analysis (flow rate, gate opening and area)

Equilibrium height

The equilibrium height can be found with the first derivative, making it equal to zero

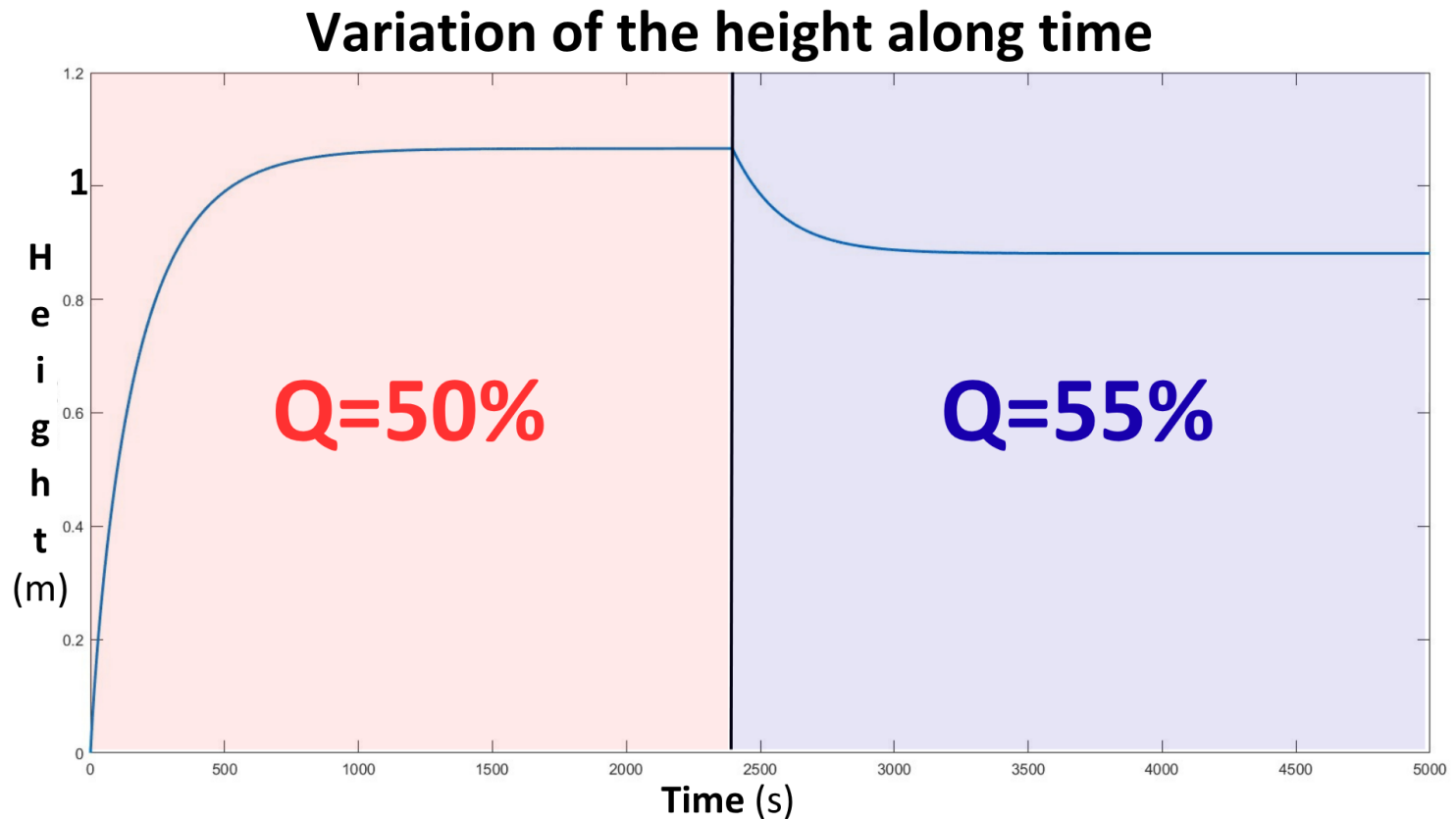
$$-u \frac{b\sqrt{2g}}{A} \sqrt{h_{eq}} + \frac{10^{-3}}{A} Q = 0 \quad \Rightarrow \quad h_{eq} = \left(\frac{10^{-3} Q}{ub\sqrt{2g}} \right)^2$$

- If Q (gate opening) increases $\rightarrow$ Equilibrium height increases
- If u (water flow) $\rightarrow$ Equilibrium height increases

NOTE: The water flow (u) and the area (A) control the speed of response, by making the first derivate bigger or smaller

System Behavior (1) – Gate

- If we change the opening of the gate, we can see the system reacting, achieving a new equilibrium water height



Second Derivative

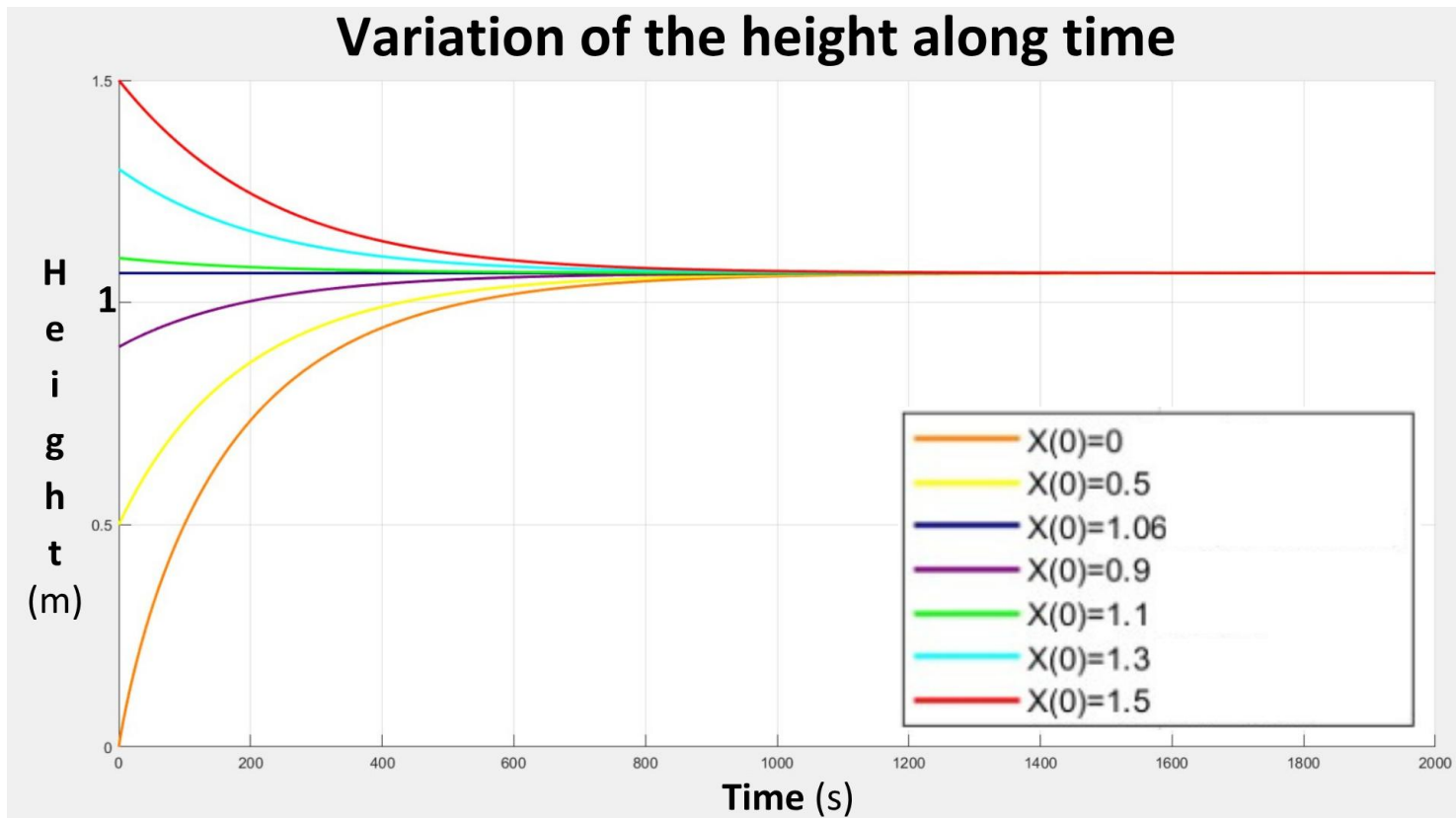
$$\frac{dh(t)}{dt} = -\frac{ub\sqrt{2gh(t)}}{A} + \frac{10^{-3}Q}{A}$$

$$\frac{dh^2}{dt^2} = -\frac{1}{2}u\frac{b\sqrt{2g}}{A\sqrt{h_{eq}}}\frac{dh}{dt}$$

Variable	Signal				
$h(t)$	0	...	$P_e = \left(\frac{10^{-3}Q}{ub\sqrt{2g}}\right)^2$	...	h_{max}
$\frac{dh}{dt}$	+	+	0	-	-
$\frac{d^2h}{dt^2}$	-	-	0	+	+
Shape	∩	∩	Inflection Point	U	U

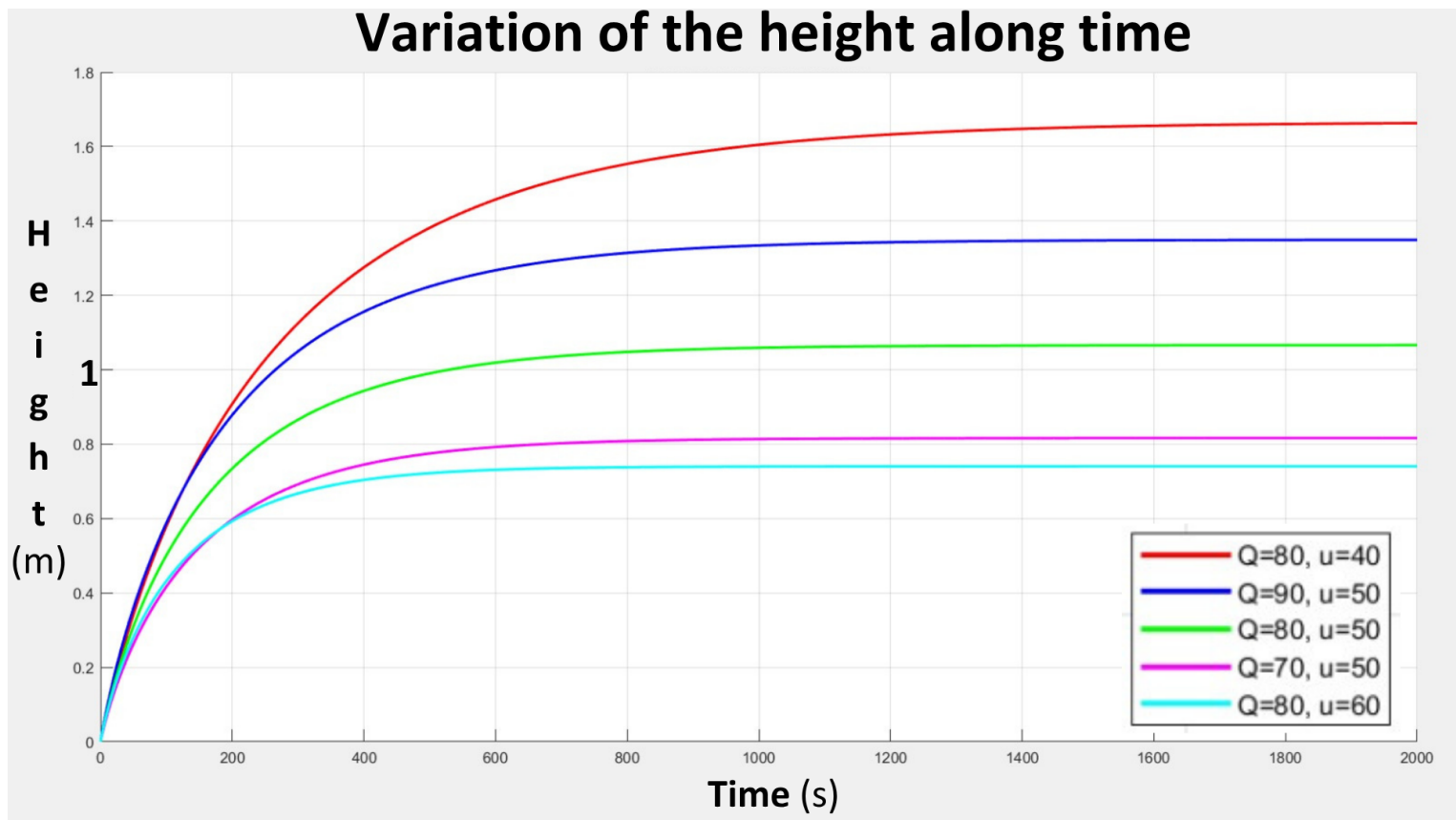
System Behavior (2) – Initial height

- If $h < \text{equilibrium height}$ $\rightarrow$ height increases until stable.
- If $h > \text{equilibrium height}$ $\rightarrow$ height decreases until stable.
- All cases converge to the same equilibrium point.



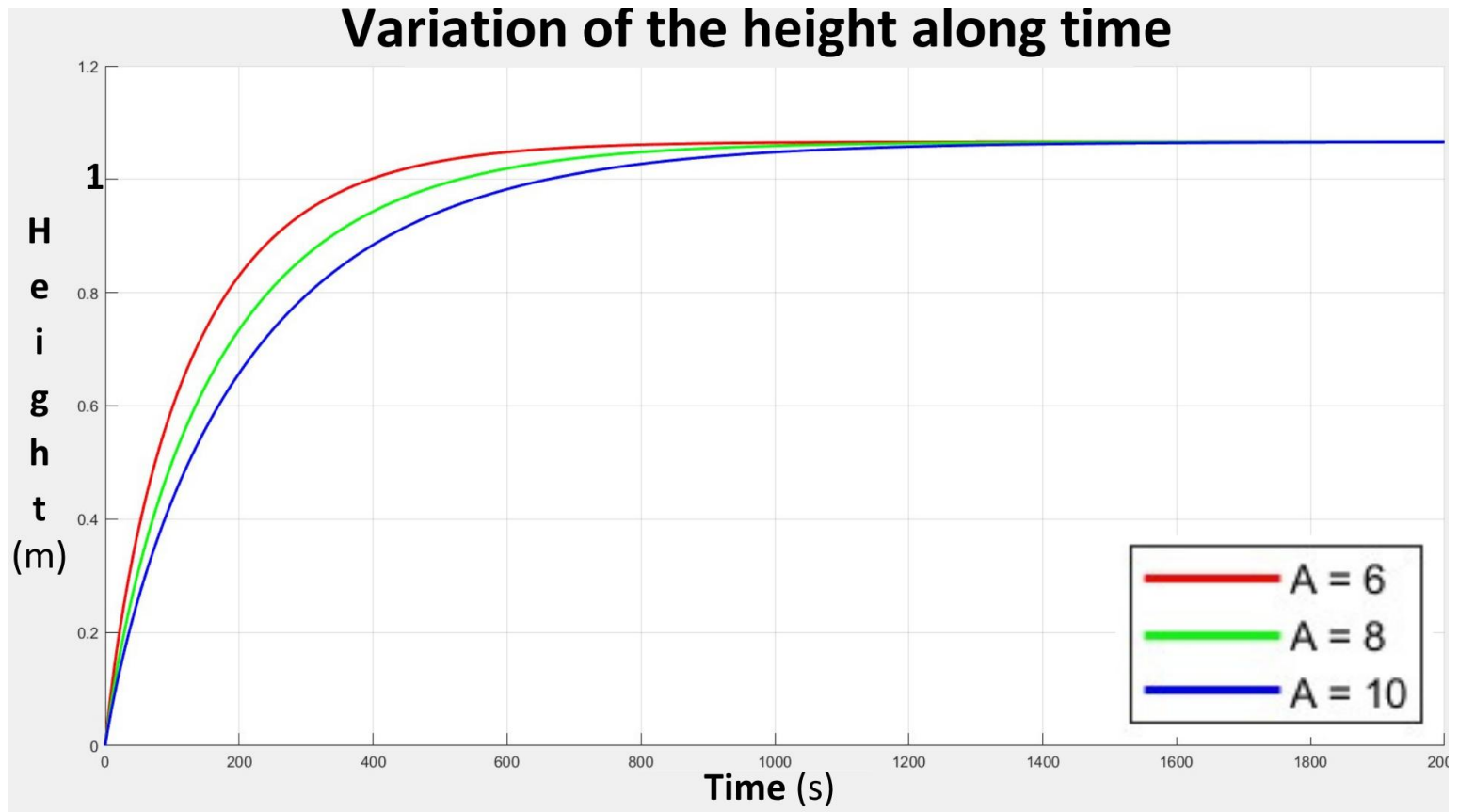
System Behavior (3) – Water flow

- Increasing the water flow (u) reduces the equilibrium height and makes the system stabilize faster.



System Behavior (4) - Area

- Increasing area (A) do not change the equilibrium height and makes the system stabilize faster.



Conclusions

- Successfully modelled water channel modulation.
- Verified stability equilibrium through MATLAB simulation.
- Demonstrated how parameters influence system response.
- Limitation: neglects nonlinear effects and real Saint-Venant complexity.

Thank you!

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